Esercizi di Fondamenti di Automatica - 2
Corso di Laurea in Ingegneria Elettronica
A.A. 2020/2021

Esercizio 1. Tracciare il diagramma di Bode delle seguenti funzioni di trasferimento:

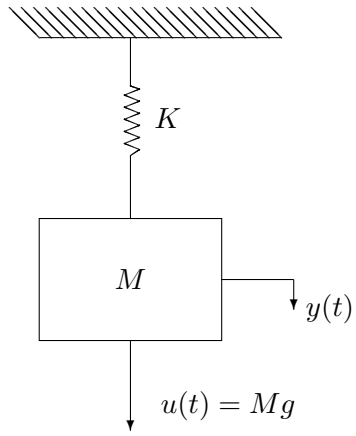
1. $W(s) = \dfrac{s}{s+1}$;
2. $W(s) = \dfrac{s-1}{s+1}$;
3. $W(s) = \dfrac{5}{(s+1)(s+5)}$;
4. $W(s) = \dfrac{s+10}{(s+0.1)(s+1)}$;
5. $W(s) = \dfrac{s-1}{s(s+10)}$;
6. $W(s) = \dfrac{s-1}{s^2}$;
7. $W(s) = 10\dfrac{s+0.1}{(s-1)(s+1)}$;
8. $W(s) = 10^3\dfrac{s(s+10)}{(s+10^2)^2}$;
9. $W(s) = 10\dfrac{s(s-10^{-1})}{(s+10)^2(s^2-1)}$;
10. $W(s) = \dfrac{(s-10)}{s^2(s+10)^2}$.

Esercizio 2. Si consideri il sistema dinamico SISO a tempo continuo descritto dalla seguente equazione differenziale:

$$\frac{d^2y(t)}{dt^2} + 100\frac{dy(t)}{dt} = \frac{du(t)}{dt} + u(t), \qquad t \in \mathbb{R}_+,$$

dove y è l'uscita e u l'ingresso. Si determini la risposta in frequenza del sistema e se ne tracci il diagramma di Bode.

Esercizio 3. Si consideri il seguente sistema meccanico



dove M rappresenta la massa del corpo sospeso, K la costante elastica della molla e indichiamo con ν il coefficiente di attrito che avversa il moto del corpo.

i) Determinare la risposta in frequenza $W(j\omega)$ del sistema;

ii) supponendo $M = 1Kg$, $K = 0.2N/m$ e $\nu = 0.01N \cdot s/m$, si tracci il diagramma di Bode di $W(j\omega)$.

Esercizio 4. Si consideri il sistema dinamico SISO a tempo continuo descritto dalla seguente equazione differenziale:

$$\frac{d^3y(t)}{dt^3} + 11\frac{d^2y(t)}{dt^2} + 10\frac{dy(t)}{dt} = 100\frac{du(t)}{dt} - u(t), \qquad t \in \mathbb{R}_+.$$

i) Si determini la risposta in frequenza del sistema e

ii) se ne tracci il diagramma di Bode.

Esercizio 5. Tracciare il diagramma di Bode delle seguenti funzioni di trasferimento:

1. $W(s) = \dfrac{s}{s^2+1}$;
2. $W(s) = \dfrac{s+1}{s^2+2s+2}$;
3. $W(s) = \dfrac{s+10}{(s+0.1)(s^2+1)}$;
4. $W(s) = \dfrac{s-1}{s(s^2+6s+25)}$;
5. $W(s) = \dfrac{s+1}{s^2+2s+9}$;

6. $W(s) = 10\dfrac{s+0.1}{(s-1)^2(s^2+1)}$.

7. $W(s) = 20\dfrac{s(s+0.1)}{(s^2+2s+9)^2}$;

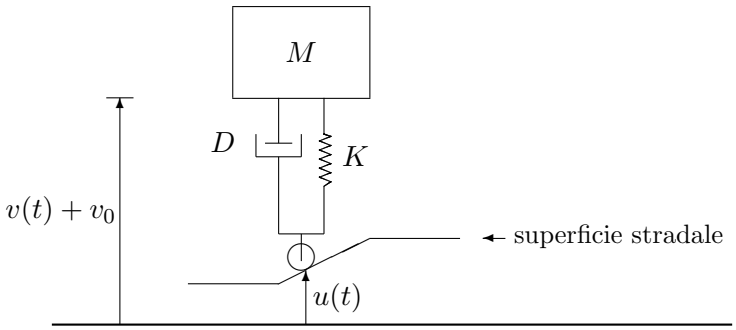
8. $W(s) = \dfrac{s+0.1}{s^2(s^2+1)(s-10)}$;

9. $W(s) = 100\dfrac{s+1}{s(s^2+1)(s-10)^2}$;

10. $W(s) = \dfrac{s^2+2s+100}{s(s^2+1)(s-10)}$;

11. $W(s) = 0.1\dfrac{s-100}{s^2(s^2-0.s+1)(s+10)}$.

Esercizio 6. [Sospensioni di un'automobile] Nel Capitolo 1 del libro abbiamo introdotto il sistema massa-molla-smorzatore e abbiamo elencato, tra le sue possibili applicazioni, la descrizione approssimata delle sospensioni di un'automobile. Vogliamo ora prendere in considerazione con maggior dettaglio questo problema. L'obiettivo delle sospensioni di un'automobile è quello di filtrare le brusche variazioni nell'andamento dell'automobile dovute alle irregolarità del terreno. In tal senso il sistema si comporta come un filtro passa-basso.



Nella figura precedente, y_0 rappresenta la distanza tra la massa M (dello chassis) dell'auto e la superficie stradale in condizioni di riposo, $y(t) + y_0$ la posizione dell'auto rispetto al riferimento di quota (indipendentemente dalle irregolarità del fondo stradale), mentre $u(t)$ rappresenta lo scostamento della posizione del fondo stradale rispetto al riferimento di quota. L'equazione descrittiva della dinamica del sistema risulta:

$$M\frac{d^2y(t)}{dt^2} + D\frac{dy(t)}{dt} + Ky(t) = Ku(t) + D\frac{du(t)}{dt},$$

dove M è la massa dell'auto, K e D sono le costanti di molla e smorzatore.

i) Si studi la stabilità BIBO del sistema.

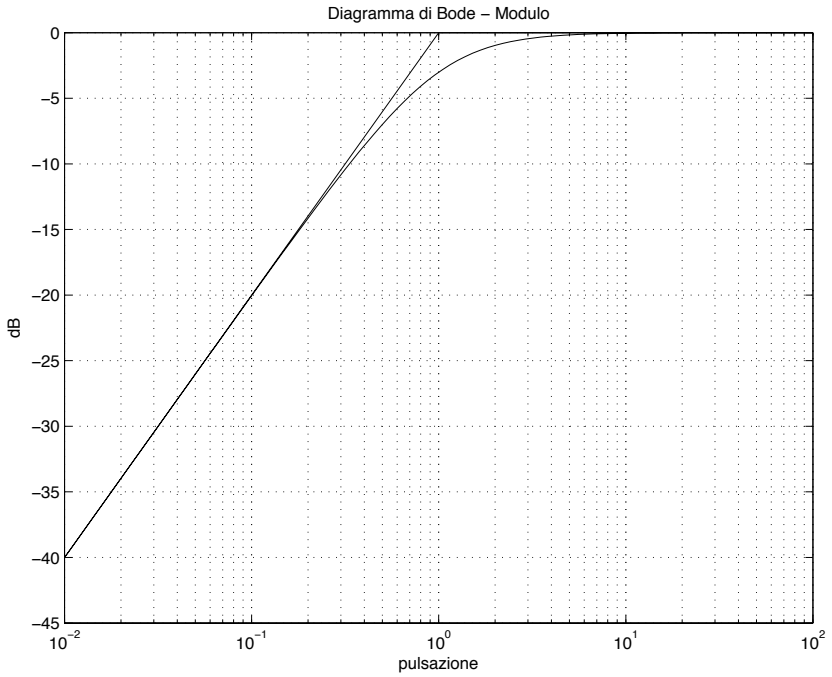
ii) Si determini la risposta in frequenza del sistema.

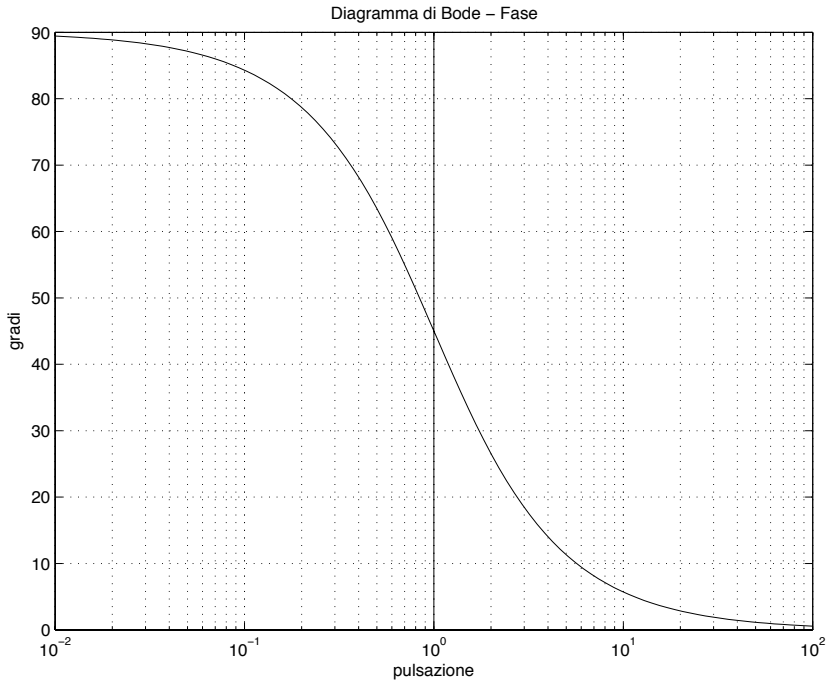
iii) Assumendo $D^2 < 4KM$, situazione tipica per i valori delle costanti in gioco in un sistema fisico, si tracci il diagramma di Bode della risposta in frequenza precedentemente determinata.

iv) Si determini la risposta al gradino $u(t) = \delta_{-1}(t)$ del sistema in esame.

Soluzioni numeriche di alcuni esercizi

Esercizio 1. 1)





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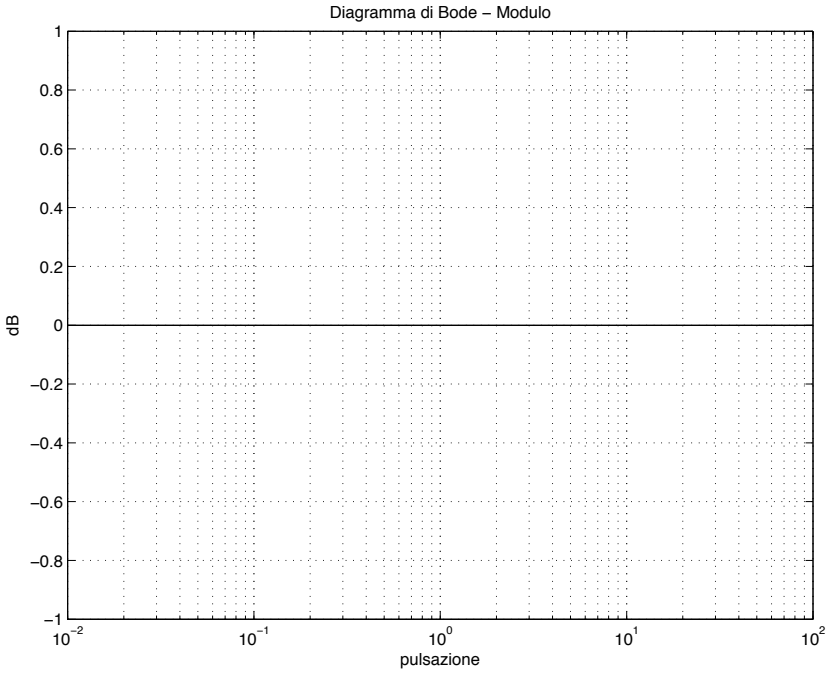


Diagramma di Bode − Modulo
dB
pulsazione

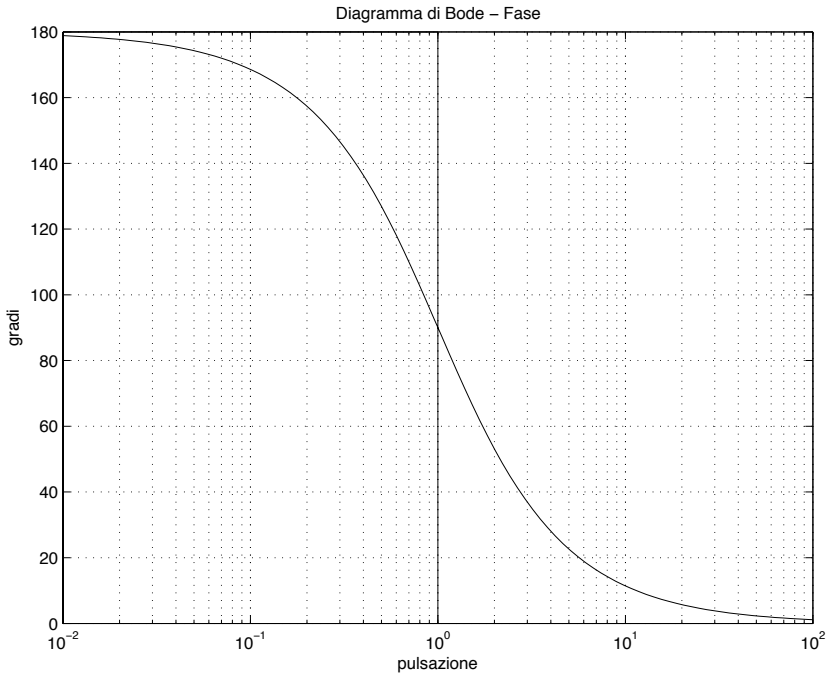


Diagramma di Bode − Fase
gradi
pulsazione

3)

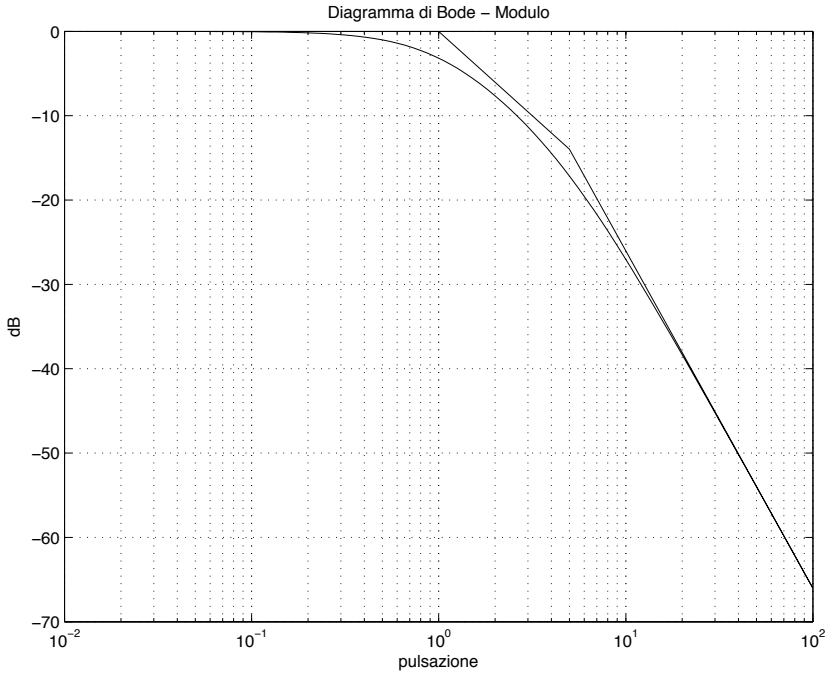


Diagramma di Bode – Modulo
dB
pulsazione
0
−10
−20
−30
−40
−50
−60
−70
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

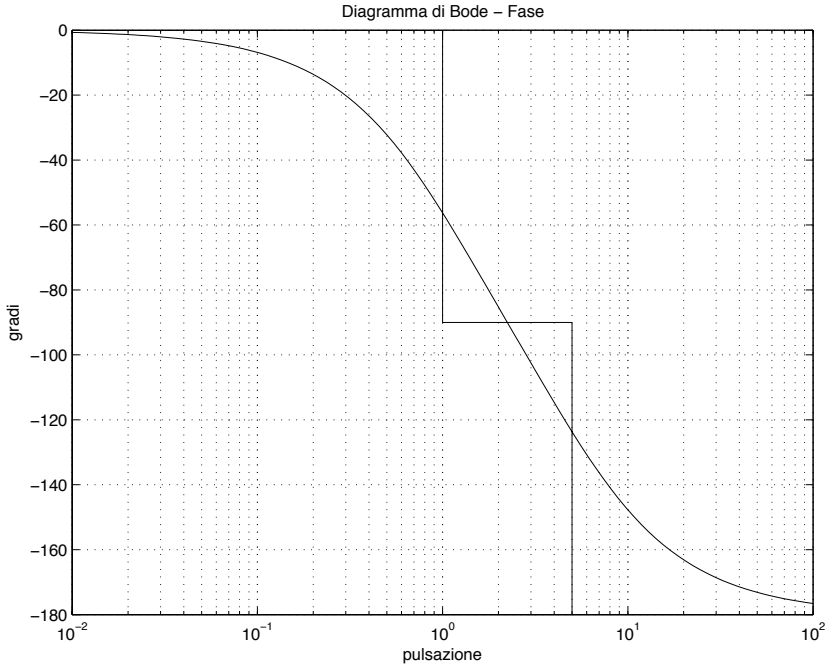


Diagramma di Bode – Fase
gradi
pulsazione
0
−20
−40
−60
−80
−100
−120
−140
−160
−180
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

4)

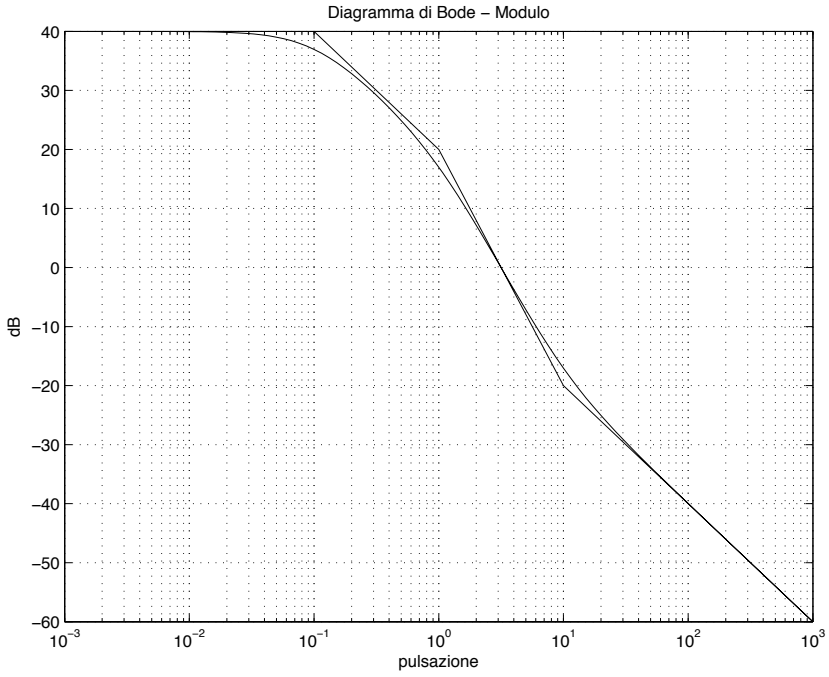


Diagramma di Bode – Modulo
dB
pulsazione
40
30
20
10
0
−10
−20
−30
−40
−50
−60
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}

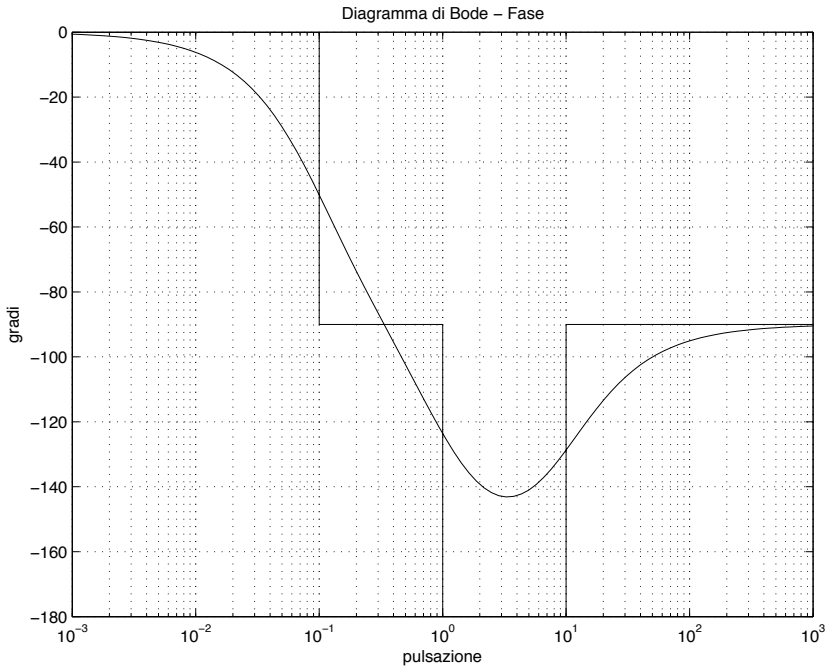


Diagramma di Bode – Fase
gradi
pulsazione
0
−20
−40
−60
−80
−100
−120
−140
−160
−180
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}

5)

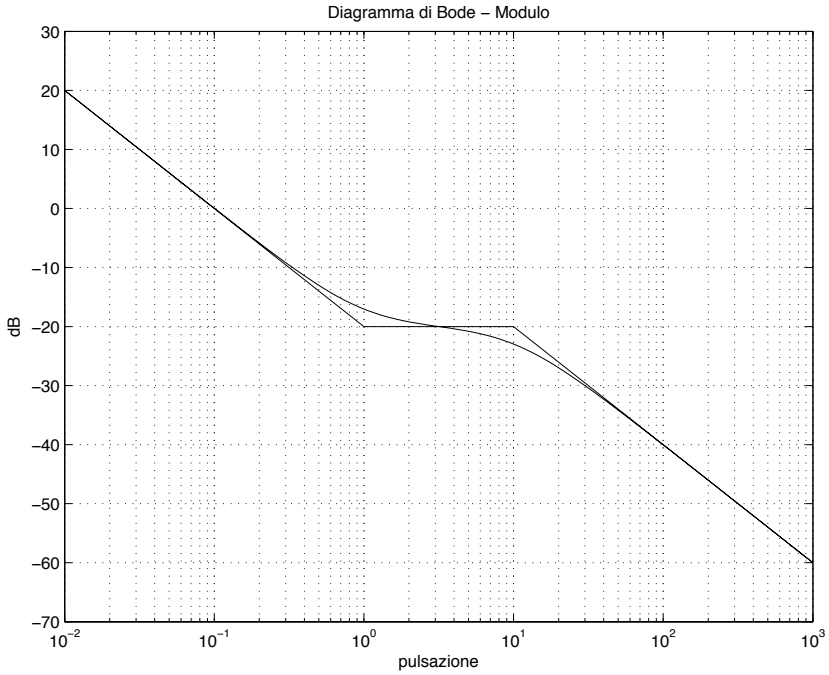


Diagramma di Bode – Modulo
dB
pulsazione
30
20
10
0
−10
−20
−30
−40
−50
−60
−70
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}

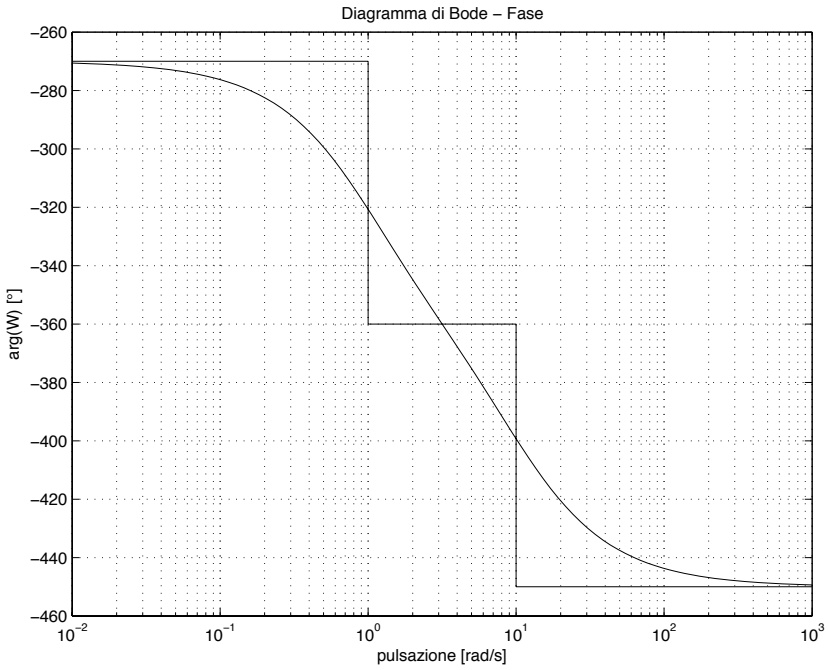


Diagramma di Bode – Fase
arg(W) [°]
pulsazione [rad/s]
−260
−280
−300
−320
−340
−360
−380
−400
−420
−440
−460
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}

6)

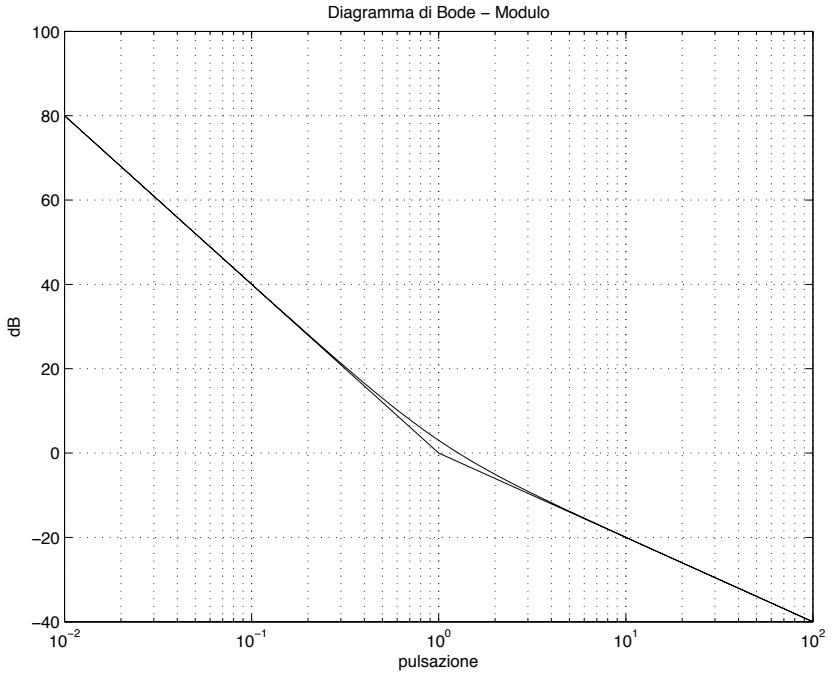


Diagramma di Bode – Modulo
dB
pulsazione
100
80
60
40
20
0
−20
−40
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

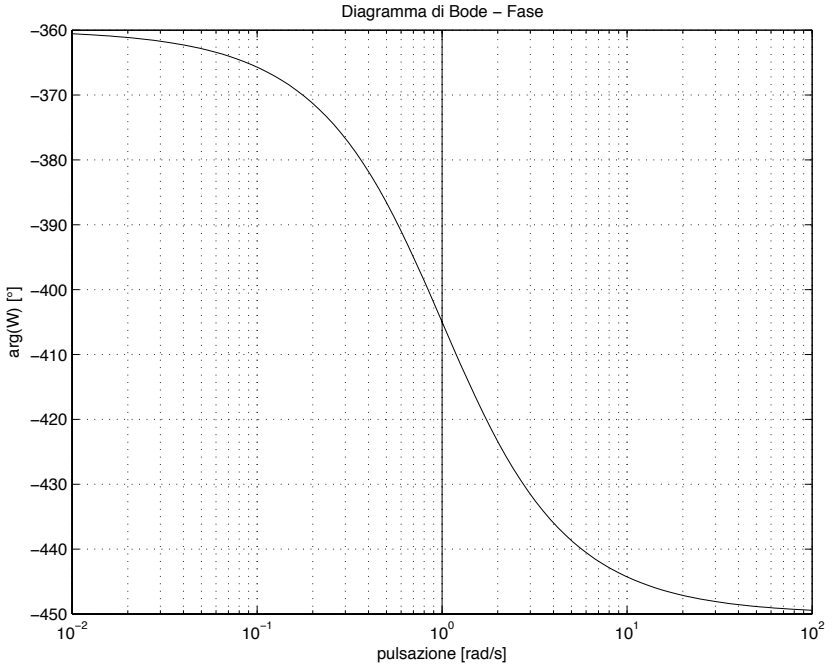


Diagramma di Bode – Fase
arg(W) [°]
pulsazione [rad/s]
−360
−370
−380
−390
−400
−410
−420
−430
−440
−450
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

7)

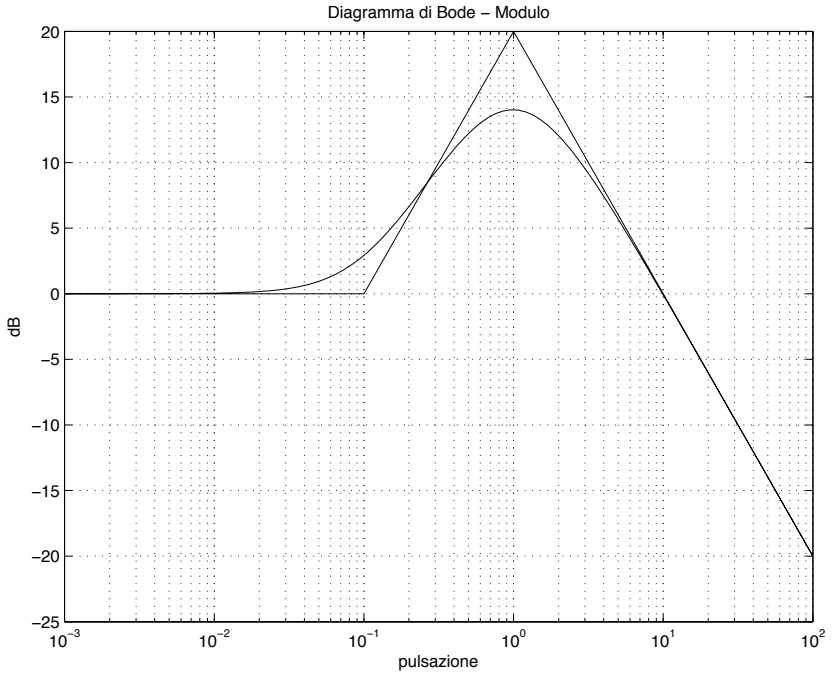


Diagramma di Bode – Modulo
dB
pulsazione
20
15
10
5
0
−5
−10
−15
−20
−25
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

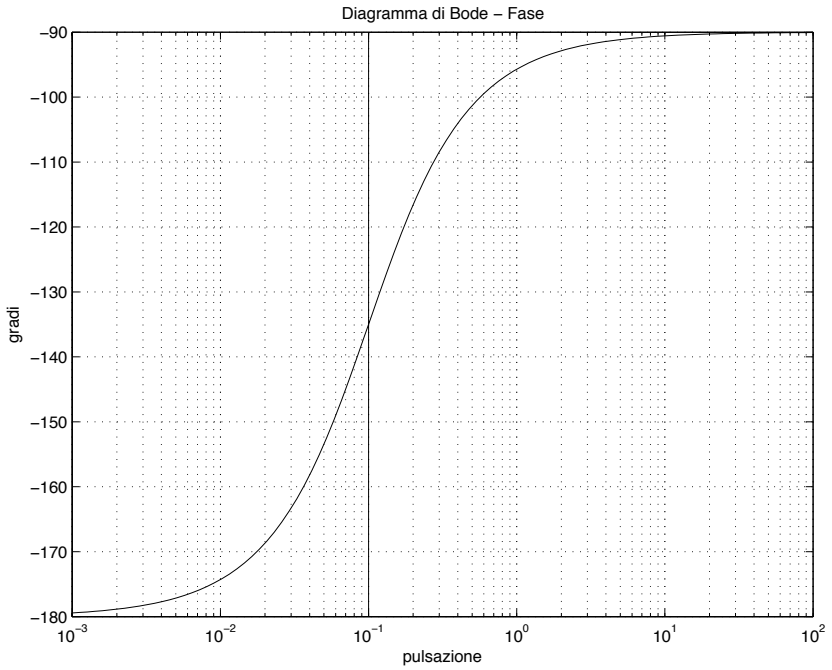


Diagramma di Bode – Fase
gradi
pulsazione
−90
−100
−110
−120
−130
−140
−150
−160
−170
−180
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

8)

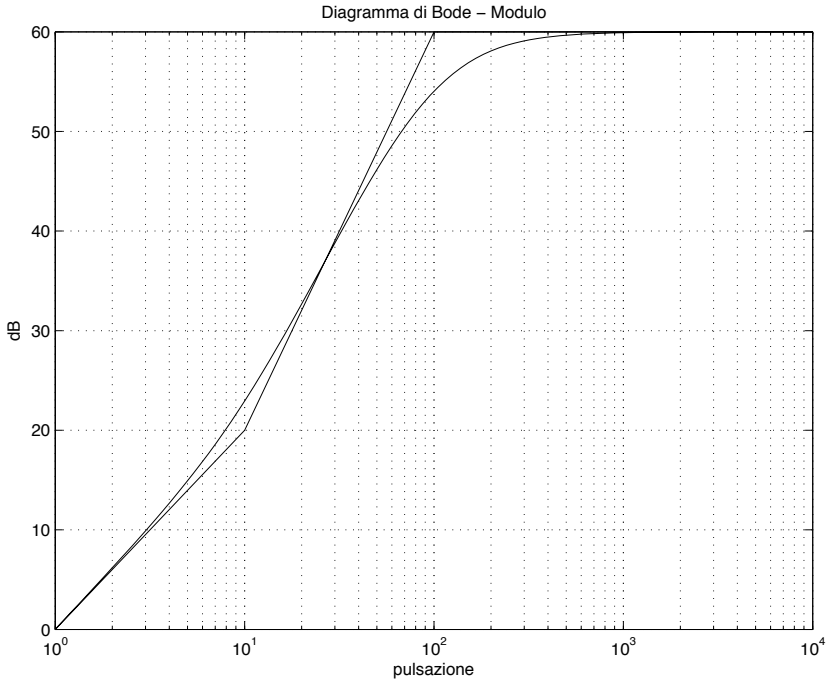


Diagramma di Bode – Modulo
dB
pulsazione
0
10
20
30
40
50
60
10^0
10^1
10^2
10^3
10^4

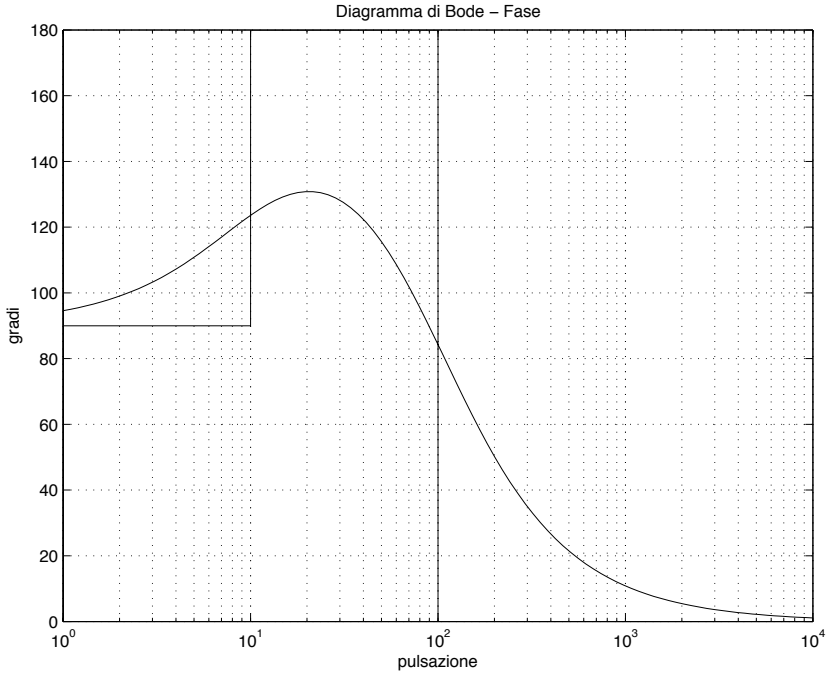
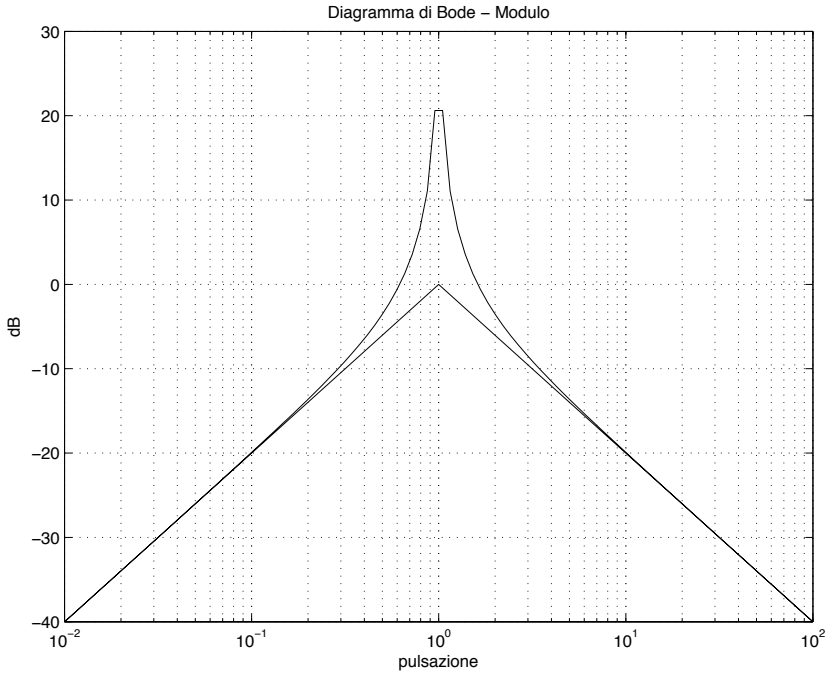
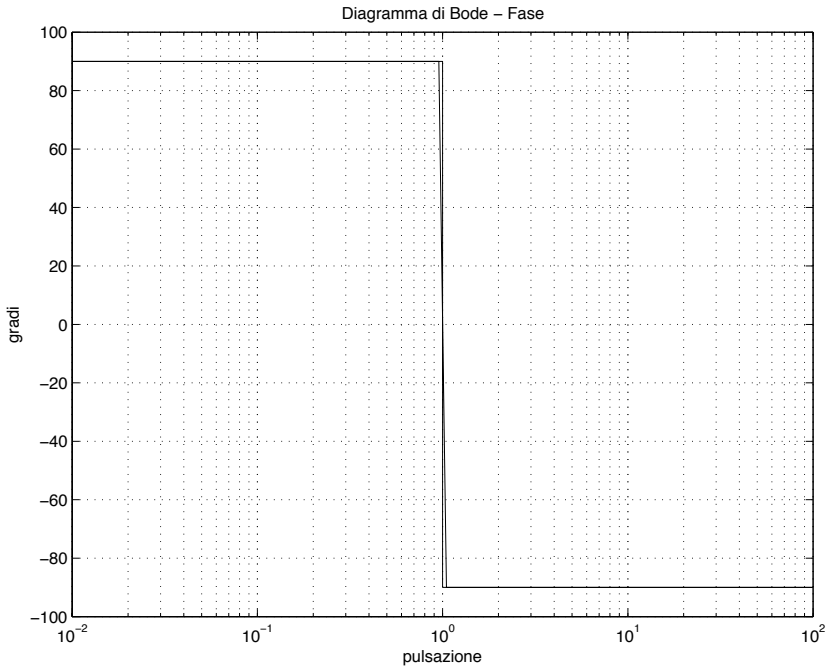


Diagramma di Bode – Fase
gradi
pulsazione
0
20
40
60
80
100
120
140
160
180
10^0
10^1
10^2
10^3
10^4

Esercizio 5. 1)





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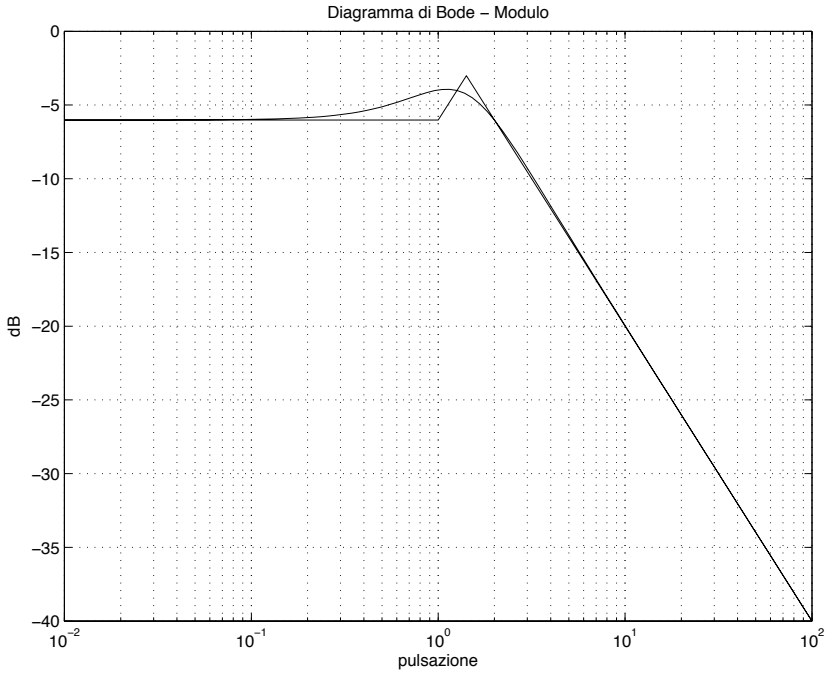
Diagramma di Bode – Modulo
dB
pulsazione
0
−5
−10
−15
−20
−25
−30
−35
−40
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

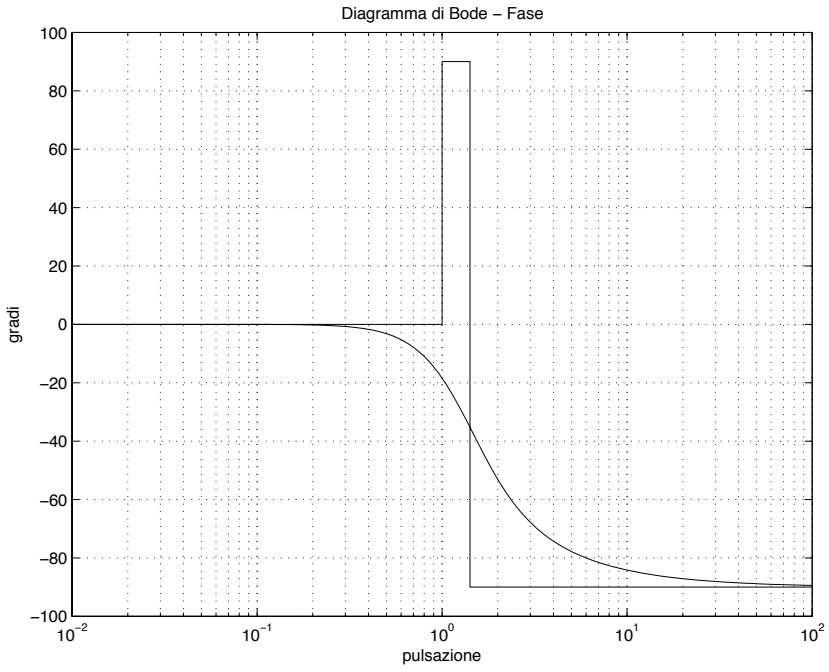
Diagramma di Bode – Fase
gradi
pulsazione
100
80
60
40
20
0
−20
−40
−60
−80
−100
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

3)

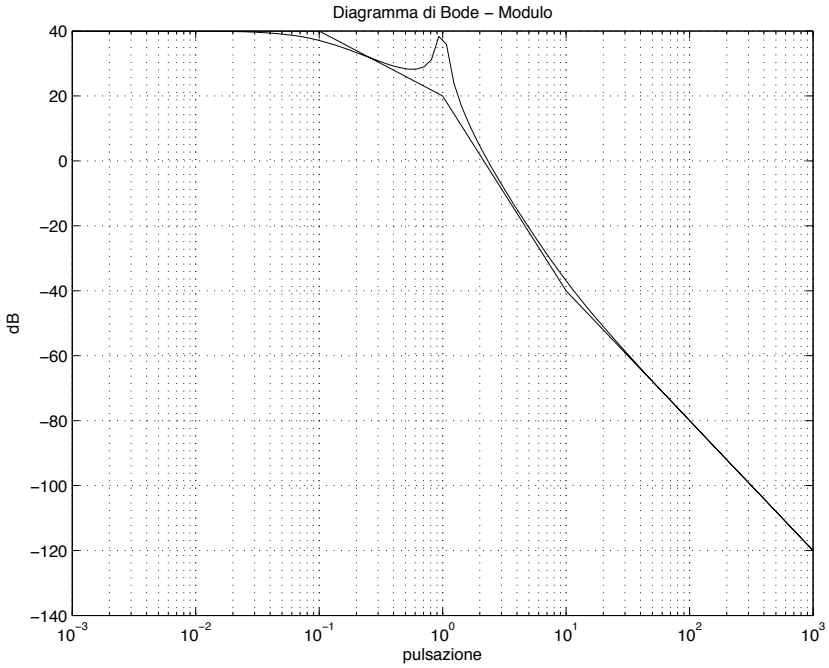
Diagramma di Bode – Modulo
dB
pulsazione
40
20
0
−20
−40
−60
−80
−100
−120
−140
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}

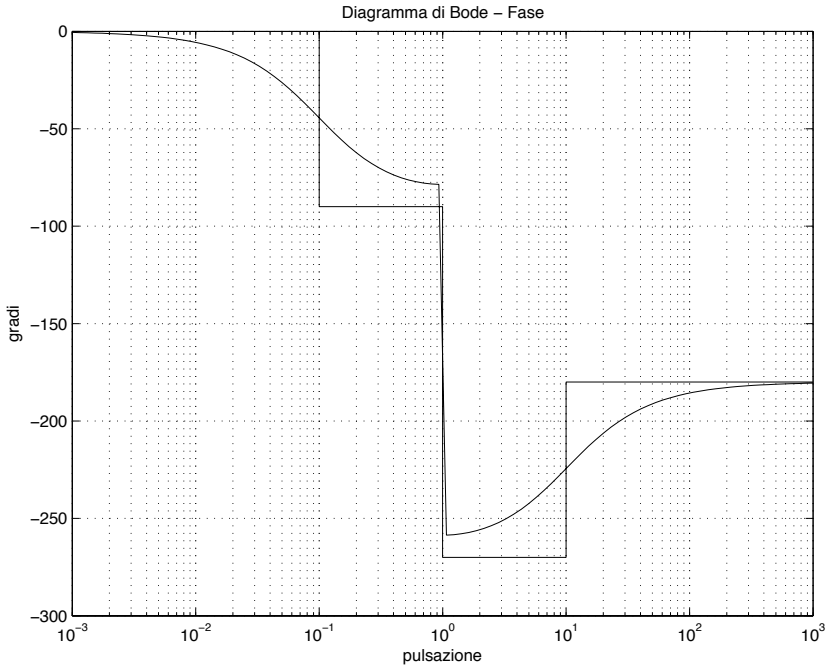
Diagramma di Bode – Fase
gradi
pulsazione
0
−50
−100
−150
−200
−250
−300
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}

4)

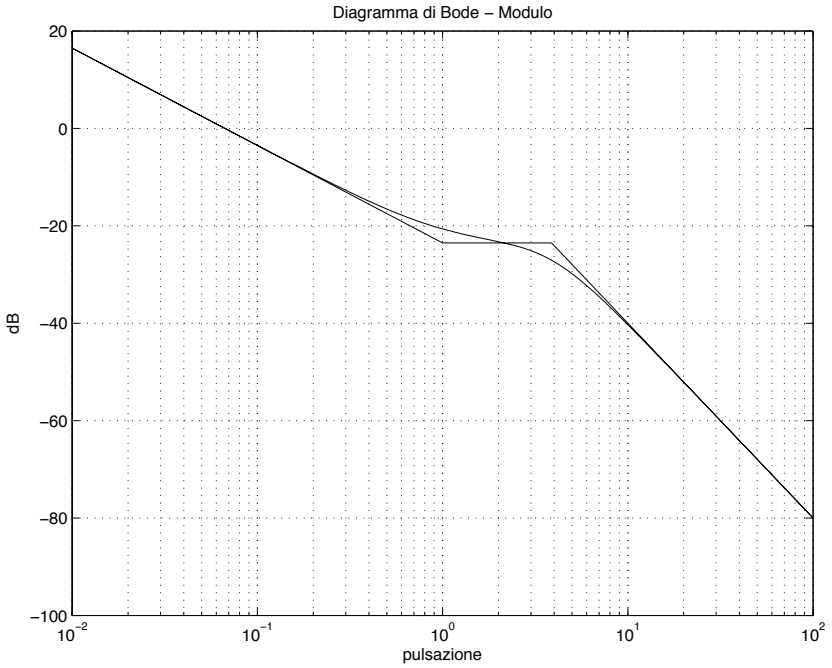
Diagramma di Bode – Modulo
dB
pulsazione
20
0
−20
−40
−60
−80
−100
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

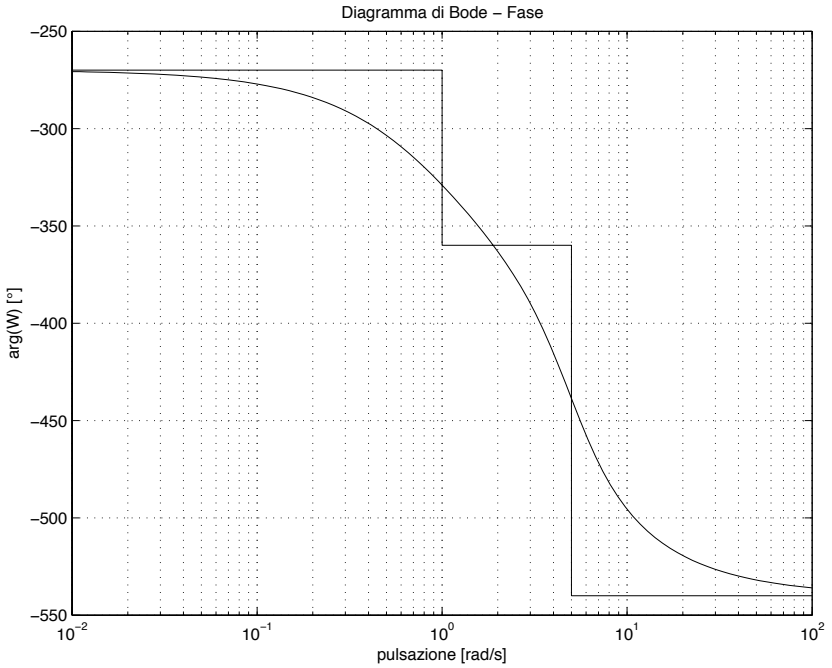
Diagramma di Bode – Fase
arg(W) [°]
pulsazione [rad/s]
−250
−300
−350
−400
−450
−500
−550
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

5)

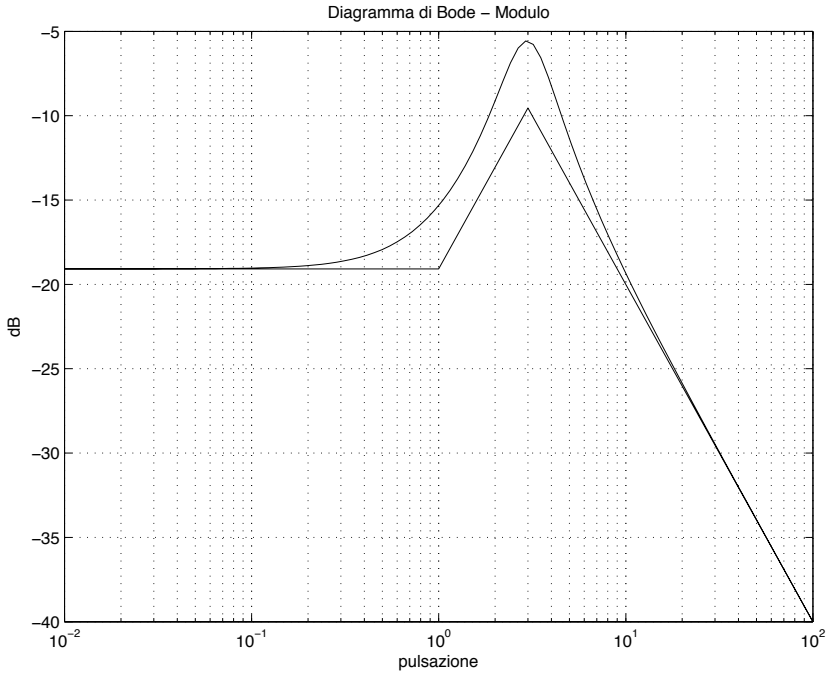
Diagramma di Bode – Modulo
dB
pulsazione
−5
−10
−15
−20
−25
−30
−35
−40
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

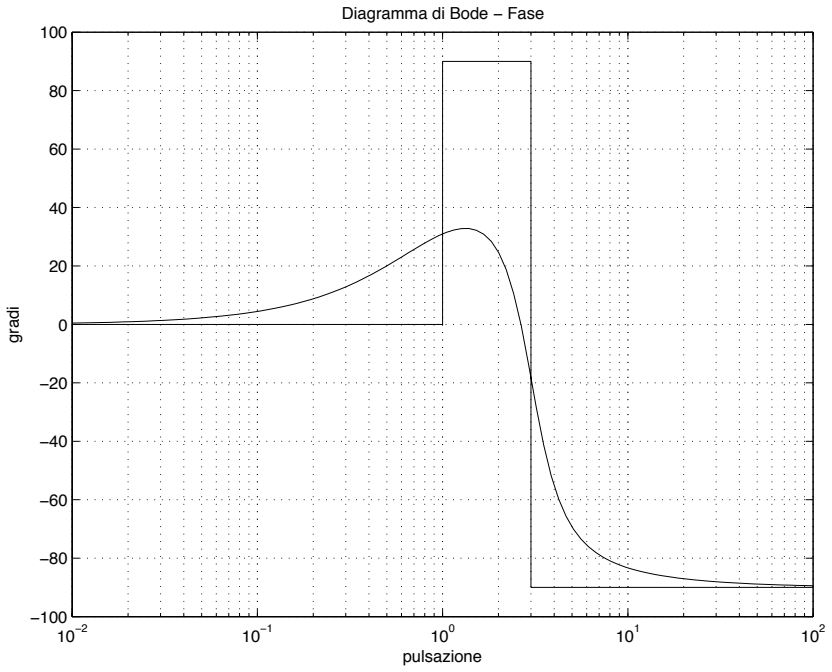
Diagramma di Bode – Fase
gradi
pulsazione
100
80
60
40
20
0
−20
−40
−60
−80
−100
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

6)

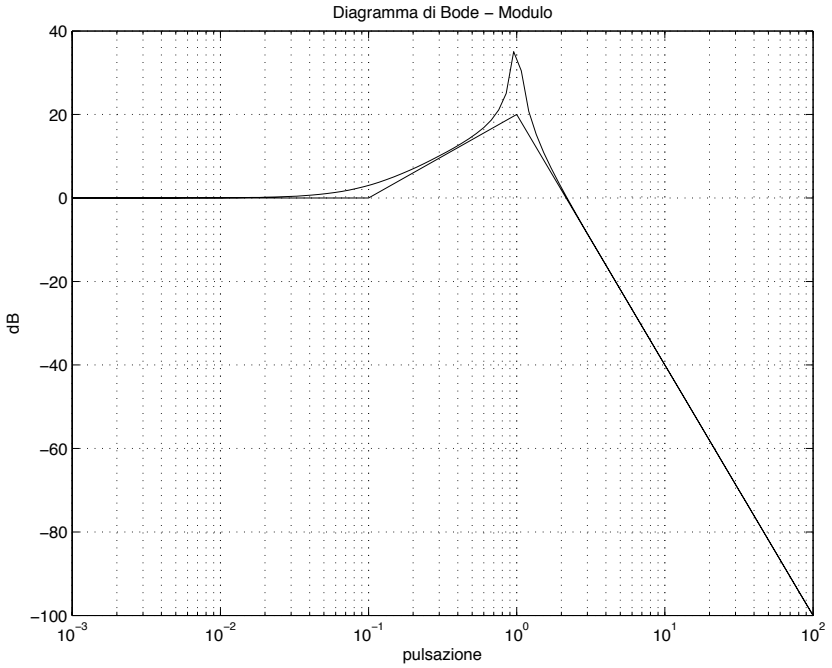
Diagramma di Bode – Modulo
dB
pulsazione
40
20
0
−20
−40
−60
−80
−100
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

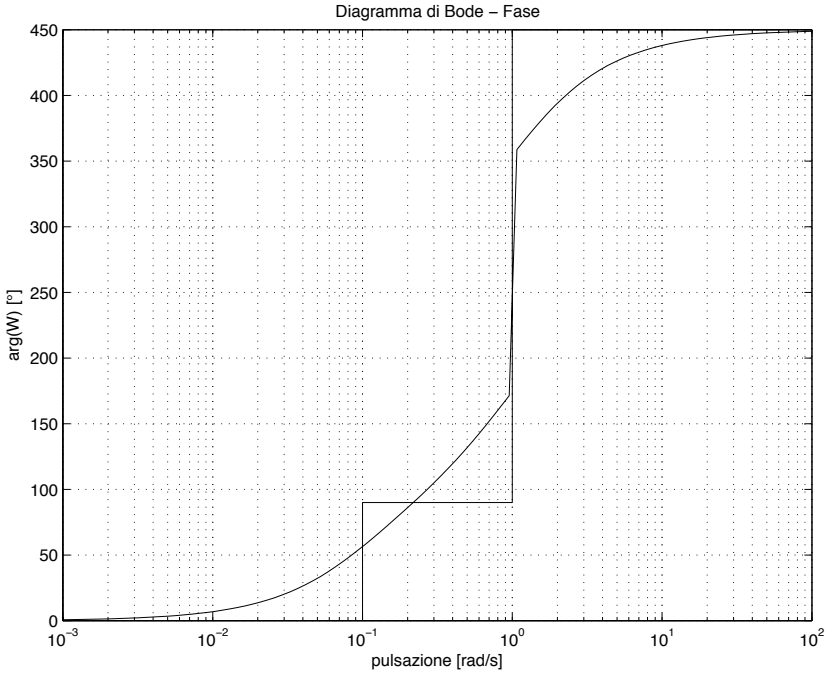
Diagramma di Bode – Fase
arg(W) [°]
pulsazione [rad/s]
450
400
350
300
250
200
150
100
50
0
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

7)

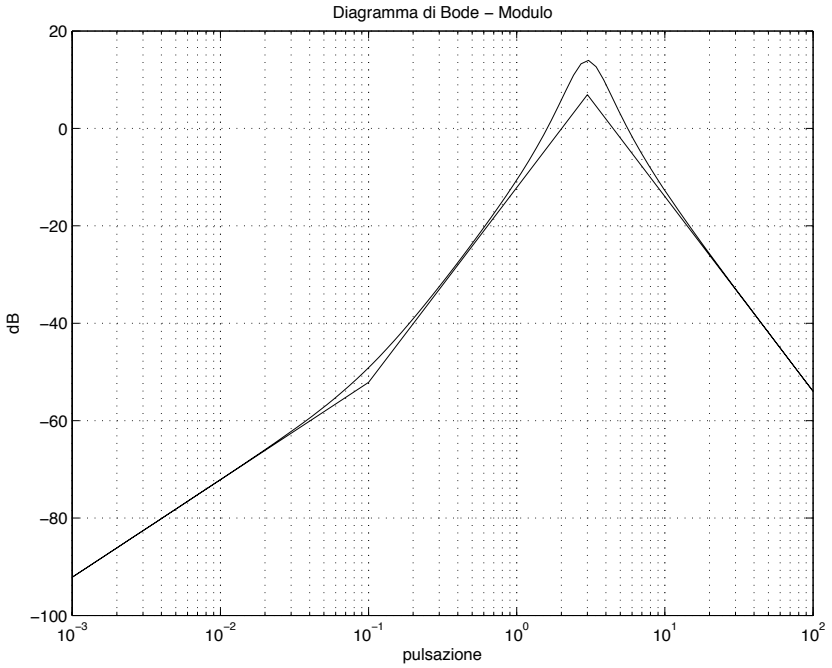
Diagramma di Bode – Modulo
dB
pulsazione
20
0
−20
−40
−60
−80
−100
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

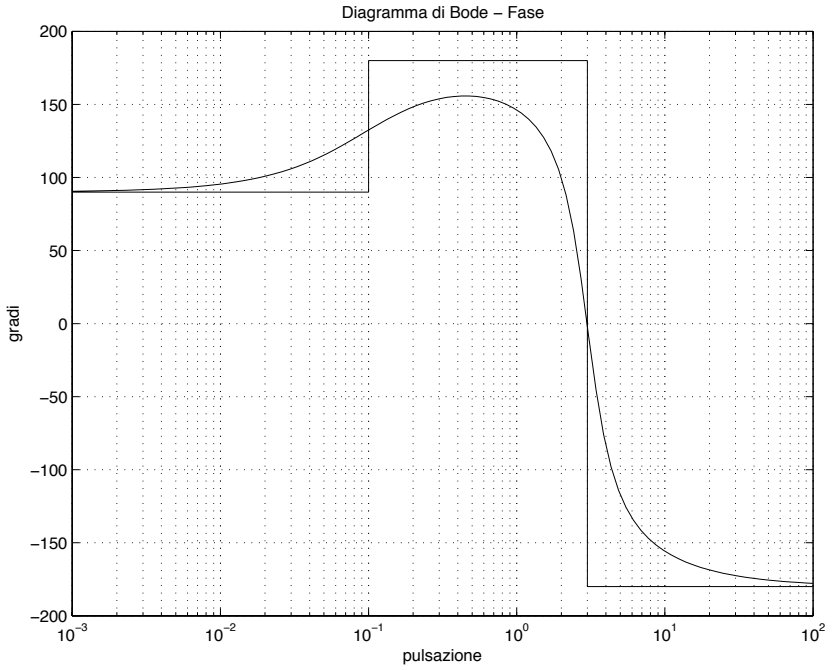
Diagramma di Bode – Fase
gradi
pulsazione
200
150
100
50
0
−50
−100
−150
−200
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

8)

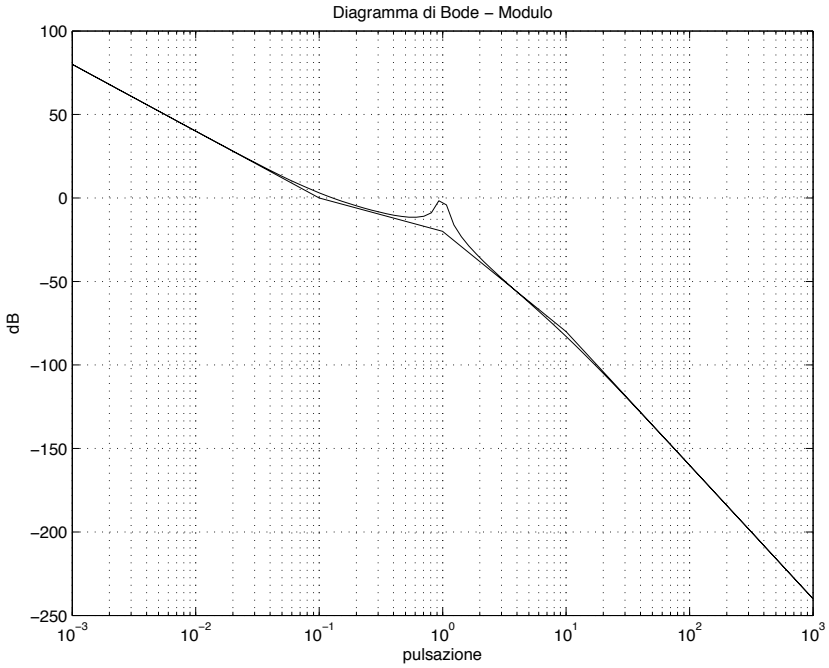
Diagramma di Bode – Modulo
dB
pulsazione

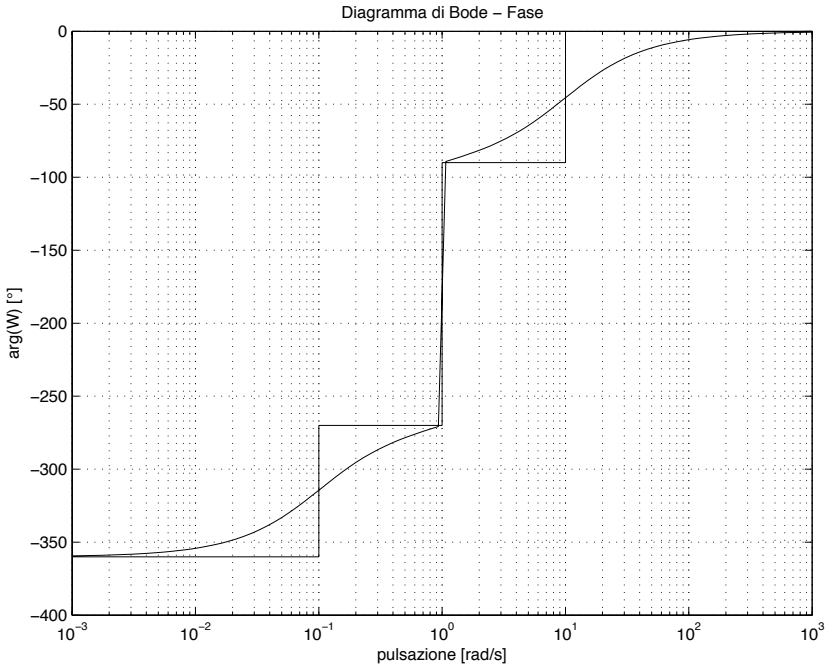
Diagramma di Bode – Fase
arg(W) [°]
pulsazione [rad/s]

9)

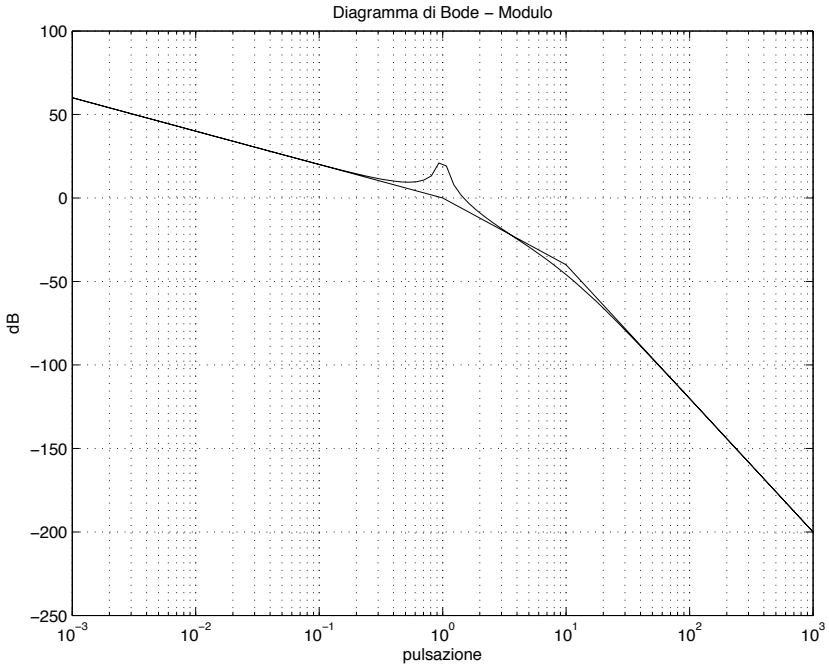
Diagramma di Bode – Modulo
dB
pulsazione

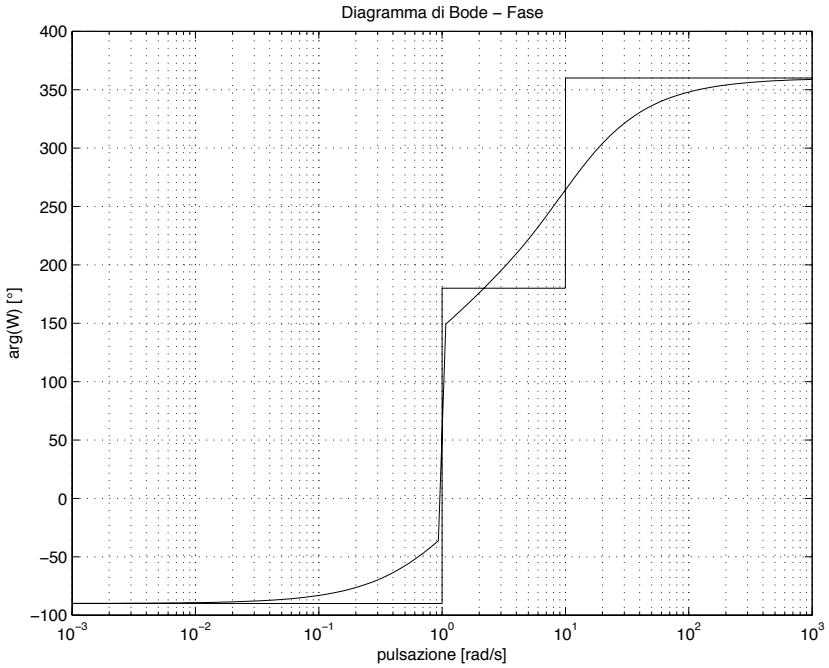
Diagramma di Bode – Fase
arg(W) [°]
pulsazione [rad/s]